1 Low-scale seesaw models (5 points)

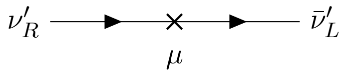
This exercise will explore low-scale variants of the seesaw mechanism. The idea will be to turn the seesaw on its head and explore symmetries to make neutrino masses small. Low-scale seesaws (sometimes referred to as double, extended, linear, or inverse seesaws) introduce a pair of neutral lepton per neutrino family (call them ν_R and ν'_R, for example). In a single-generation scenario, the neutrino mass matrix becomes

$$H_{\text{int}} = \Big(|\nu_L\rangle \; |\bar{\nu}_L\rangle \; |\bar{\nu}'_L\rangle \Big) \begin{pmatrix} 0 & m_D & \varepsilon \\ m_D & \mu' & M \\ \varepsilon & M & \mu \end{pmatrix} \begin{pmatrix} \langle \bar{\nu}_R | \\ \langle \nu_R | \\ \langle \nu'_R | \end{pmatrix} + \text{h.c.} \tag{1.1}$$

Recall: ν_L is the usual (interacting) left-handed neutrino and ν_R and ν'_R are additional (sterile) neutral leptons. Because of CPT, they come with corresponding conjugate states $|\nu_L\rangle \leftrightarrow |\bar{\nu}_R\rangle$, $|\nu_R\rangle \leftrightarrow |\bar{\nu}_L\rangle$, and $|\nu'_R\rangle \leftrightarrow |\bar{\nu}'_L\rangle$. The state $|\bar{\nu}_R\rangle$ is interacting (as an antineutrino) and $\bar{\nu}_L$ and $\bar{\nu}'_L$ are sterile.

1. The quantum numbers of the neutral leptons are $L(\nu_L) = +1$, $L(\nu_R) = +1$, and $L(\nu'_R) = -1$. Out of all the parameters in the matrix above, which ones violate lepton number?

By inspecting the quantum numbers of each mass term, we can deduce that μ, μ', and ϵ violate lepton number. Take μ for instance.



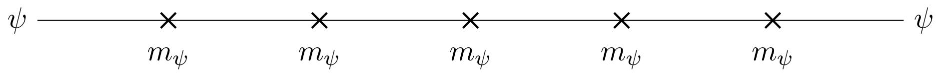
where we interpret the amplitude for ν'_R to convert to $\bar{\nu}'_R$ as

$$\mathcal{A}_{\nu'_R \to \bar{\nu}'_L} = \langle \bar{\nu}'_L | \, H_{\text{int}} \, |\nu'_R\rangle = \langle \bar{\nu}'_L | \Big(\mu \, |\bar{\nu}_L\rangle \, \langle \nu'_R | + \dots \Big) \, |\nu'_R\rangle = \mu$$

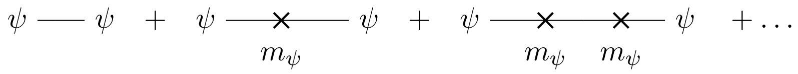
where all other terms go to zero. Clearly, the initial and the final states have opposite lepton number, so this term violates lepton number by two units. Similarly, one can show that μ' and ε also violate lepton number.

Note that we are free to assign lepton number to the sterile state ν'_R as we please, since it does not interact with the SM. Had we picked $L(\nu'_R) = +1$ instead, then μ', μ, and M would violate lepton number instead and ε would conserve it.

Why can we think of these mass terms as amplitudes? Take a fermion Dirac mass term between ψ_L and ψ_R. It turns out that if we sum all possible mass insertions as this fermion propagates, we recover the propagator of a massive particle. Take



The propagator of a mass fermion is simply $\frac{1}{\not{p}}$, which we can insert in chains of fermion mass terms. Indeed, there are an infinite number of such fermion chains,



Summing all these contributions, we get a geometric series

$$\frac{1}{\not{p}} + \frac{1}{\not{p}} m_\psi \frac{1}{\not{p}} + \frac{1}{\not{p}} m_\psi \frac{1}{\not{p}} m_\psi \frac{1}{\not{p}} + \cdots = \frac{1}{\not{p}} \sum_{n=0}^{\infty} \left(\frac{m_\psi}{\not{p}} \right)^n = \frac{1}{\not{p} - m_\psi},$$

which is indeed the propagator of a massive fermion. Strictly speaking, this series only converges for $|m_\psi/\not{p}| < 1$, but the final expression is valid more generally by analytic continuation.

You may ask: how come we can insert arbitrary number of masses in our diagrams if they change chirality? Shouldn't the incoming and outgoing states change depending on the number of mass insertions? The answer is that this "derivation" glosses over many details, but one can indeed keep track of all the chiralities with more bookkeeping. For instance, with a fermion basis $\psi = (\psi_L, \psi_R)$, one can write the propagator as a 2×2 matrix in chirality space,

$$S_F(p) = \begin{pmatrix} \frac{\not{p}}{p^2 - m_\psi^2} & \frac{m_\psi}{p^2 - m_\psi^2} \\ \frac{m_\psi}{p^2 - m_\psi^2} & \frac{\not{p}}{p^2 - m_\psi^2} \end{pmatrix},$$

where the off-diagonal terms change chirality (mass insertions) and the diagonal terms preserve chirality (kinetic terms). This matrix can be derived by summing all possible insertions of mass and kinetic terms while keeping track of chirality, generalizing what we did above.

The same logic applies to larger fermion matrices and Majorana mass terms as well, but just as the discussion involving chirality, this will involve more book-keeping.

2. Calculate the determinant of this matrix. What happens when all the lepton-number-violating (LNV) parameters go to zero? What does this imply for light neutrino masses? (Hint: recall that the determinant is the product of the eigenvalues.).

det$(M) = 2\epsilon M m_D - \epsilon^2\mu' - \mu m_D^2$. When all the LNV parameters go to zero, this determinant vanishes! This means that at least one of the eigenvalues is zero (recall that det$(M) = \Pi_i \lambda_i$ and Tr$(M) = \Sigma_i \lambda_i$). So, when lepton number is conserved, the model is no good for generating light neutrino masses. Note that in the standard seesaw, the LNV parameter (μ) is meant to be large.

3. Keeping all these lepton-number-violating parameters zero and assuming the Dirac mass m_D is much smaller than all other parameters, diagonalize the matrix above and analyze what happens with the spectrum of massive particles. What kind of particles do you find?

   By diagonalizing this matrix, we get $m_1 = 0$, $m_{2,3} = \mp\sqrt{M^2 + m_D^2} \simeq \mp M + \frac{m_D^2}{2M}$. Note that this is a Dirac fermion! There are two degenerate (chiral/Weyl/2-component) fermions, which will make up one single Dirac particle, just like in the case of the electron or muons, etc. When $m_D \ll M$, this is easy to understand, as the 2×2 sub-block mass matrix looks like the usual Dirac fermion mass matrix. The difference is that we have an additional unpaired fermion, namely the light neutrino.

The lesson here is: the seesaw generates Majorana neutrino masses by breaking lepton number. With 3 fermions, if lepton number is conserved, at least one neutrino remains massless and the other two pair into a Dirac fermion (in the same way that Dirac neutrinos arise from two Weyl fermions with lepton number conserved, except that in this case, there is an odd number of fermions).

4. Turn on only a single LNV parameter, for example $\mu \neq 0$, at a time and diagonalize the matrix assuming $m_D \ll$ LNV parameters $\ll$ anything else. (you may use Mathematica, Wolfram Alpha, or Python, but you don't have to). You can assume all parameters are real and positive.

   How do the light neutrino masses depend on the LNV parameters in this case? To explain the smallness of neutrino masses, do you expect lepton number to be a good or a badly broken symmetry in these scenarios?
   Hint: the question above should convince you that μ should be a small parameter to get neutrino masses to be small. So, expand in $m_D/M \ll 1$ as well as in $\mu/M \ll 1$. The general solution is too messy. Here is some pseudo-code for MATHEMATICA on how to do this (there are many ways to do this, of course):

```
M = {{0, mD, 0}, {mD, 0, M}, {0, M, mu}};
MdiagExp = (Series[
Eigenvalues[MISS,
 Cubics -> True] /. {mu -> t*rmu*M,
```

```
 mD -> t*rd*M}, {t, 0, 3}] // PowerExpand //
Normal) /. t -> 1 // Simplify
MdiagExp /. {rd -> mD/M, rmu -> mu/M}
```

Above I expanded to third order since the light neutrino mass only shows up at this order, but for the two heavy masses, all we need is the zeroth order term. **Super-Hint**: Btw, this last statement should allow you to get to the answer without using any computer algebra at all. All you need is $\det\{M\} = -\mu m_D^2 = \lambda_1\lambda_2\lambda_3$ and the answer to the previous question.

When only μ is non-zero, we get

$$m_1 = \frac{m_D^2}{M^2}\mu, \quad m_{2,3} = \mp M + \frac{m_D^2}{2M} + \dots \tag{1.2}$$

Note the heavy masses are basically unchanged from before since they are dominated by M, the largest scale in the problem. You can see that when $\mu \to 0$, the light neutrino mass goes to zero as well.

You could have obtained this result without much work by noting that $\det(M) = -\mu m_D^2 = m_1 m_2 m_3 \simeq -m_1 M^2 \implies m_1 = \frac{m_D^2}{M^2}\mu$.

Diagonalizing the 3×3 matrix is quite the challenge, but the important part is the expression for the lightest mass,

$$m_\nu = \frac{m_D^2\mu + \varepsilon^2\mu' - 2M\varepsilon m_D}{M^2}. \tag{1.3}$$

Again, when all LNV parameters go to zero, this goes to zero.

If the only non-zero LNV parameter is ε, then neutrino masses are proportional:

$$m_\nu \simeq -2\frac{m_D}{M}\varepsilon. \tag{1.4}$$

This is called the **linear** seesaw since neutrino masses depend linearly on all new mass parameters.

Yet another interesting case is when only μ' is non-zero. In this case, we get $m_\nu = 0$. No reason to worry though, since this result is subject to quantum corrections at the one-loop level. In that case, neutrino masses are generated proportional to μ'.

5. What kind of particle are the heavy neutral leptons in our spectrum? Dirac, quasi-Dirac, or Majorana fermions?

> The heavy masses are roughly $m_{2,3} \sim \mp M$: this means that the mass eigenstates 2 and 3 combine into a Quasi-Dirac fermion. The two (Weyl) fermions have almost the same mass, but are split by the scale of LNV parameters. So, in the limit of lepton number (quasi-)preservation, the heavy neutrinos become more and more Dirac like.

> You can show that the (small) mixing angles are still given by things of the order of m_D/M. This is great because the small LNV parameters that can reduce the size of neutrino masses *do not* reduce the size of the mixing between active and heavy neutrinos. The great advantage of low-scale seesaws is that the neutrino mass mechanism is a lot more testable. They are also “natural” in the technical sense: neutrino masses are small because there is a symmetry that is enhanced when they go to zero. This is the opposite of what happens in the usual seesaw: there the neutrino masses are small because of a hierarchy of scales.

2 Heavy neutrino decay and detection (5 points)

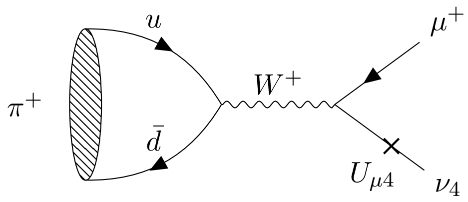
Let us now look at how you could search for a single heavy neutral lepton (HNL) in an accelerator experiment. Let’s call this HNL ν_4 with mass m_4 and assume it mixes predominantly with the muon neutrino flavor via a mixing angle $|U_{\mu 4}|^2$. In other words, every time a muon neutrino is produced, there is a probability $|U_{\mu 4}|^2$ that it is actually a heavy neutrino ν_4 instead. Diagramatically, this can be represented as:

$$\nu_\mu \longrightarrow \underset{U_{\mu 4}}{\times} \longrightarrow \nu_4$$

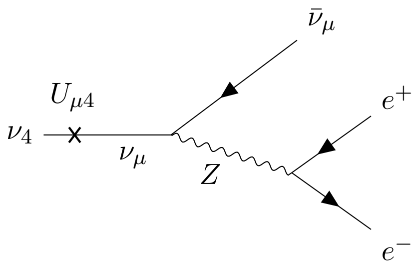
This exercise will guide you through an estimate of the signal event rate for HNLs produced by pion decay-at-rest, which subsequently decay inside a large-volume detector far away. Consider a spallation source with $\mathcal{O}(\mathrm{GeV})$ protons hitting a dense target. The number of pions produced at this source is related to the number of protons on target by $N_\pi \sim 10\% \times N_{\mathrm{POT}}$. As soon as they are produced, they lose energy and stop inside the target, decaying at rest via $\pi^+ \to \mu^+ \nu_\mu$. Typical accelerators produce $N_{\mathrm{POT}} \sim 10^{23}$ protons on target per year.

1. Using the mixing line drawn above, draw a Feynman diagram for the production of HNLs from pion decay $\pi^+ \to \mu^+ \nu_4$. Now also draw a decay diagram for the HNL to decay via $\nu_4 \to \nu_\mu e^+ e^-$.

The production diagram is shown below:



The decay diagram is very similar to muon decay, but can only proceed via a Z boson:



Note how we used the mixing in both diagrams to swap a muon neutrino for a heavy neutrino, allowing to interact through the weak interaction.

2. The branching ratio for $\pi^+ \to \mu^+ \nu_4$ is approximately

$$\mathcal{B}(\pi^+ \to \mu^+ \nu_4) \sim |U_{\mu 4}|^2 \mathcal{B}(\pi^+ \to \mu^+ \nu_\mu), \tag{2.1}$$

where $\mathcal{B}(\pi^+ \to \mu^+ \nu_\mu) \simeq 100\%$. Here, $|U_{\mu 4}|^2$ is the mixing angle of the heavy neutrino with the muon neutrinos [1].

Using that branching and the number of pions produced per POT and the fact that the decay to HNLs is isotropic, what is the flux of HNLs at a distance L from the target per year of operation? Account for the lifetime of the HNL by multiplying this flux by the probability for ν_4 to survive the distance L without decaying.

Now estimate the number of HNLs that do decay within a region of volume $V = \lambda^3$ located at a distance L from the source. Assume $\lambda \ll L$. Hint: recall that the probability for a particle to decay within a time t is $P_{\text{dec}} = 1 - e^{-t/\tau_{\text{LAB}}}$. To survive a time t, the probability is $1 - P_{\text{dec}}$. Recall, $\tau_{\text{LAB}} = 1/\Gamma_{\text{LAB}}$.

[1] Note this is analogous to θ^2 in exercise 1, except that it is now phrased in a 4-neutrino mixing picture and makes use of the relation $|\nu_\alpha\rangle = \sum_i^4 U^*_{\alpha i} |\nu_i\rangle$

HNLs will be produced by pion decay. Since pions decay predominantly as $\pi^+ \to \mu^+\nu_\mu$, we can just swap $\nu_\mu \to N$ by paying the price of $|U_{\mu 4}|^2$. For N_{HNLs} Given $N_\pi \sim 10\% \times N_{\text{POT}}$, we then have $N_{\text{HNLs}} \sim 10\% \times N_{\text{POT}} \times |U_{\mu 4}|^2$. Since we want the flux, we divide by the area of the sphere centered around the source (note the neutrino and HNL emission is isotropic!),

$$\Phi_{\text{HNLs}} = \frac{N_{\text{HNLs}}}{4\pi L^2}. \tag{2.2}$$

A particle decay survival probability is given by $P = e^{-\tau/\tau_L}$ with τ_L the lab-frame lifetime. In terms of the decay with, $\tau_L = \gamma/\Gamma$, where γ is the boost of the HNL. Now, we want the probability to decay inside the detector after a distance L. First, the particle must survive a distance L, so $P_{\text{survival}} = e^{-\Gamma_{\text{N}} L/\beta\gamma}$, where β is the velocity of the HNL, $\beta = v/c$. Now we want it to decay within a linear segment inside the detector. Let's assume it is of size λ, so $P_{\text{decay}} = 1 - e^{-\Gamma_N \lambda/\beta\gamma}$. Overall, the probability for a HNL to make it to the detector and decay inside of it is just the product of these two things: $P_{\text{decay}} \times P_{\text{survival}}$.

3. Knowing that the muon lifetime is determined by the decay rate $\Gamma = G_F^2 m_\mu^5/192\pi^3$, estimate the analogous decay rate for $\nu_4 \to \nu_\mu e^+ e^+$. Hint: compare the Feynman diagrams and use the mixing suppression as shown in the beginning of this question.

Well, no trick here, just again, take $\Gamma_{\nu_4} \sim |U_{\mu 4}|^2 G_F^2 m_\mu^5/192\pi^3$. In principle, since I did not specify what kind of outgoing neutrino we have in the decay, you could also write $\Gamma_{\nu_4} \sim (|U_{e4}|^2 + |U_{\mu 4}|^2 + |U_{\tau 4}|^2) G_F^2 m_\mu^5/192\pi^3$, corresponding to the sum of $\nu_4 \to \nu_e e^+ e^-$, $\nu_4 \to \nu_\mu e^+ e^-$, and $\nu_4 \to \nu_\tau e^+ e^-$. This is not exactly correct since we are neglecting factors of order 1 and the fact that the decay $\nu_4 \to \nu_e e^+ e^-$ can proceed through either W or Z, but it is good enough for our estimates.

4. Putting all of this together, write down a formula for the number of HNL decays within this detector of size V. You can assume the HNL decays only via $\nu_4 \to \nu e^+ e^-$ with a rate Γ_{ν_4}. Expand this formula assuming $\Gamma_{\nu_4} L$ and $\Gamma_{\nu_4} \lambda$ are very small. How does the signal event rate depend on the decay rate Γ_{ν_4}?

First we want the number of HNLs that pass through our detector. We can estimate it assuming the area of the detector is roughly λ^2. In that case, the total number of HNL decays is

$$N_{\text{tot}} = \frac{N_{\text{HNLs}}}{4\pi L^2} \times \lambda^2 \times P_{\text{decay}} \times P_{\text{survival}} \sim \frac{N_{\text{HNLs}}}{4\pi L^2} \times \lambda^2 \times \frac{\Gamma_N \lambda}{\beta\gamma}, \tag{2.3}$$

where we expanded the exponentials to first order. Note that this expression is actually proportional to the volume of the detector, $V \sim \lambda^3$. It is also proportional to the signal rate Γ_N. If you consider another decay rate, then we would need to include an additional branching ratio factor in the whole formula. You can convince yourself that in the end, since both exponentials only depend on the total decay rate $\Gamma_N = \Gamma_{\nu e^+ e^-} + \Gamma_{\nu\nu\nu}$, in the long-lifetime regime ($\Gamma_N L/\gamma \ll 1$ and $\Gamma_N \lambda/\gamma \ll 1$), the total lifetime drops out of the formula and the signal is only proportional to the decay rate you are interested in ($\Gamma_{\nu e^+ e^-}$ in our case).